Midterm Review

Weeks 1–6: Themes and Key Results

IEOR E4576 — Data-Driven Models in Finance — Fall 2026

1 Week 1: Foundations of Financial Data — Exchanges, Market Structure, Probability

1.1 Limit order book and matching

- CLOB: best bid/ask, spread = ask − bid, mid = (bid + ask)/2, depth, market impact.
- Spread components: order processing, inventory risk, adverse selection (only the last is informational).
- Matching (price priority first), then: price–time/FIFO (equities, ES = S&P 500 E-mini future, ZN = 10-year Treasury note future), pro rata (SOFR futures), modified pro rata, price–size. Pro rata fill = $Q \cdot q_i / \sum_j q_j$; encourages size inflation.
- Implied pricing: implied-in (spreads from outrights), implied-out (outrights from spreads).

1.2 Order types

- Market, limit, stop, stop-limit; TIF: Day, GTC, GTD; IOC (fill what you can, cancel rest), FOK (all now or nothing), AON (no partials, may rest).
- Iceberg (loses time priority on refresh), pegged, hidden, MOO/MOC/LOC.
- Off-book: EFP, block trades.

1.3 Venues and products by asset class

- **Equities:** ∼16 exchanges + 30+ ATS/dark pools; Reg NMS, NBBO, SIP vs. direct feeds; closing auction ≈10% of volume.
- **Commodity futures:** one venue per contract (no fragmentation). NYMEX: CL, NG, RB, HO; COMEX: GC, SI, HG; CBOT: ZC, ZS, ZW; CME: LE, HE; ICE US: softs.

1.4 Futures term structure (preview of Week 2)

- Contango $F > S$ (storage dominates); backwardation $F < S$ (convenience yield dominates).
- Roll return $(F_{t,T_1} - F_{t,T_2})/F_{t,T_2}$: positive in backwardation, negative in contango for a long.

1.5 Elementary probability

- $E[X+Y] = E[X] + E[Y]$ always; $\mathrm{Var}(X+Y) = \mathrm{Var}\,X + \mathrm{Var}\,Y + 2\,\mathrm{Cov}(X,Y)$; $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}\,X$.
- Tower law $E[X] = E[E[X|Y]]$; total variance $\mathrm{Var}\,X = E[\mathrm{Var}(X|Y)] + \mathrm{Var}(E[X|Y])$.
- Bayes: $P(B_i|A) = P(A|B_i)P(B_i)/\sum_j P(A|B_j)P(B_j)$ — base rates dominate (rare disease, DNA match).
- MGF $M(t) = E[e^{tX}]$, $E[X^n] = M^{(n)}(0)$, $M_{X+Y} = M_X M_Y$ (indep.). Bernoulli $1-p+pe^t$; Poisson $e^{\lambda(e^t-1)}$; exponential $\lambda/(\lambda - t)$; normal $e^{\mu t + \sigma^2 t^2/2}$.
- Characteristic function $\phi(t) = E[e^{itX}]$ always exists (Cauchy has no moments); inversion underlies FFT option pricing.
- Moments: finite k-th ⇒ finite $j < k$; Pareto moments exist only for $k < \alpha$. CLT needs $\sigma^2 < \infty$.
- Combinatorics: hypergeometric (Keno), poker hands out of $\binom{52}{5}$; discrete uniform mean $(n+1)/2$, var $(n^2-1)/12$.
- *(Optional)* Markov chains / first-step analysis: $E_0 = 1 + \frac{5}{6}E_0 + \frac{1}{6}E_1$; fundamental matrix $N = (I-Q)^{-1}$, expected times $N\mathbf{1}$. Pattern self-overlap sets waiting time ((2,3): 36 rolls; (1,1): 42).

1.6 Stylized microstructure facts and spreads

- Inverse cubic tails $P(|r| > x) \sim x^{-3}$; volatility clustering; no return autocorrelation (negative at tick level: bid–ask bounce); square-root impact $\Delta P \sim \sigma\sqrt{Q/V}$; book imbalance $(V_b - V_a)/(V_b + V_a)$; U-shaped intraday volume.
- Crack spread 3:2:1 with unit conversion (42 gal/bbl): $S = 3\,CL - 84\,RB - 42\,HO$; mean-reverting (refiners adjust utilization).

2 Week 2: Working with Futures Data — Spreads, Rolls, Continuous Contracts

2.1 Contract mechanics

- Month codes F G H J K M N Q U V X Z (Jan–Dec); e.g. CLH24 = Mar 2024 WTI (1,000 bbl).
- Settlement price is the official mark.
- Open interest is a stock (contracts outstanding); volume is a flow. Opening trades raise OI, closing trades lower it, transfers leave it unchanged.

2.2 Continuous contracts: four decisions

- Decide: roll trigger, roll window n, weight schedule, price adjustment.
- Triggers: time-based (e.g. 5 days before LTD; deterministic), volume-based (use *prior*-day volume to avoid

look-ahead), OI-based (lagged).
- Blended price $P_t = w_t P_t^A + (1 - w_t) P_t^B$: single switch ($n = 1$), linear, volume-weighted. A window spreads the phantom roll return over n days; it doesn't remove it.
- Roll 5–10 days before LTD; volatility explodes near expiry (NG, April 2020 negative WTI).

2.3 Price adjustment
- Additive (back-adjust): $\delta = P_\tau^B - P_\tau^A$, $P_t^{adj} = P_t + \delta$ for $t < \tau$. Preserves \$ differences; can go negative; % returns wrong.
- Ratio: $\rho = P_\tau^B / P_\tau^A$, $P_t^{adj} = \rho P_t$. Preserves % returns, stays positive; distorts \$ P&L.
- No method preserves both. Levels are conventions, returns are data: store raw prices + roll schedule.

2.4 Margin and clearing
- Cost of carry $F = Se^{(r+u-y)(T-t)}$; clearinghouse novates every trade.
- Initial margin $\approx k\,\sigma_{1d}\sqrt{h}\times$ notional; maintenance $\approx 90\%$ of IM. Margin call at MM restores to *IM*.
- Variation margin $\mathrm{VM}_t = qm(F_t - F_{t-1})$ telescopes to $qm(F_T - F_{t_0})$: total is path-independent, cash timing is not. Margining turns counterparty risk into liquidity risk. SPAN gives spread credits.

2.5 Commitments of Traders (COT)
- As of Tuesday, released Friday 3:30pm ET. Legacy: Commercial, Non-Commercial, Non-Reportable. Disaggregated: Producer/Merchant, Swap Dealer, Managed Money, Other. TFF for financials.
- OI identity: $\mathrm{OI} = CM_L + NC_L + NC_X + NR_L = CM_S + NC_S + NC_X + NR_S$.
- Hedging pressure $(CM_S - CM_L)/(CM_S + CM_L)$; COT index $(MM_{net} - \min_{3y})/(\max_{3y} - \min_{3y})$.
- Strong contemporaneous correlation with returns, little predictive IC. Pitfalls: align to *release* date (Friday), overlapping windows inflate t.

2.6 Seasonal calendar spreads
- Notation NG[H][J0]: long Mar, short Apr same year; align years on trading days to LTD. One observation per year.
- Well-known spreads (don't need to memorize): NG[H][J0] "widow-maker" (winter risk premium); ZC[U][Z0] old/new crop; ZW[K][N0] harvest; HE[V][Z0] cash-settled.
- Physically delivered contracts price seasonality gradually; cash-settled ones converge only near settlement.

2.7 Corporate actions and point-in-time data (continues in Week 3)
- Split n:1: divide earlier prices by n. Dividend factor $f = (P_{t_d-1} - D)/P_{t_d-1}$; cumulative factor compounds backward.
- Total return $R = (P_t + D_t)/P_{t-1} - 1$. Knowledge vs. effective vs. as-of dates; look-ahead and survivorship bias.
- Holidays: forward fill deflates volatility, interpolation fabricates data; drop non-common dates.

3 Week 3: Equity Data

3.1 Equity venues
- Primary market (IPOs); secondary: 2nd market = exchanges (lit, feed NBBO), 3rd market = OTC dealer networks (FINRA; OTCQX/QB/Pink), 4th market = ATSs/dark pools (Reg ATS).
- Dark pools: no pre-trade transparency, fills at midpoint or better, block-oriented; ∼15–18% of US volume.
- IEX (speed bump, exchange since 2016); inverted (taker–maker) pricing.

3.2 Order matching and order types (cont'd from Week 1)
- Price–time, price–size, pro rata, modified pro rata; IOC/FOK; iceberg, pegged, non-routable; MOC/LOC.

3.3 Stylized facts of returns
- No autocorrelation of r_t; fat tails (excess kurtosis 3–10, tail index 3–4, Student-t fits better).
- Volatility clustering: $\mathrm{Corr}(|r_t|, |r_{t-k}|) > 0$, slow decay — uncorrelated ≠ independent.
- Aggregational Gaussianity; leverage effect; Epps effect (correlation falls as sampling frequency rises).

3.4 Identifiers and databases
- Ticker, CUSIP (9 char: 6 issuer, 2 issue, 1 check; CRSP/IBES store 8), ISIN, SEDOL, FIGI, CIK.
- CRSP: PERMNO (security), PERMCO (company); Compustat: GVKEY; IBES ticker; RavenPack entity ID.
- CRSP–Compustat link: `ccmxpf_linktable` (linktype L*, linkprim C/P); keep linkdt ≤ date ≤ linkenddt.
- Pitfalls: share classes, stale IDs, ticker reuse — link on permanent IDs *with dates*.

3.5 Corporate actions in CRSP (cont'd from Week 2)
- FACPR $= (s(t) - s(t'))/s(t')$: 2-for-1 → 1, 3-for-2 → 0.5; pre-split price $\div(1 + \mathrm{FACPR})$.
- CFACPR covers splits only; dividends enter via RET. Negative PRC = bid/ask average (use |prc|). Use RET, not price ratios, on event days.

3.6 Point-in-time universes and correlation structure

- Select the universe with information at the sample start; today's winners ⇒ survivorship/look-ahead bias (Big-10 index: huge overstatement).
- Track by PERMCO, weight by previous-day cap.
- Effective number of bets $N_{\text{eff}} = (\sum \lambda_i)^2 / \sum \lambda_i^2$: equities few bets (high common factor), futures more. SE of correlation $\approx 1/\sqrt{T}$.
- CAPM and APT introduced at the end of Week 3 (continued in Week 4).

4 Week 4: Equity Valuation Models; Data Workflow and Storage

4.1 CAPM and APT (cont'd from Week 3)

- CAPM: $E[R_i] - r_f = \beta_i(E[R_m] - r_f)$, $\beta_i = \text{Cov}(R_i, R_m)/\text{Var}(R_m)$. Test: $\alpha_i = 0$ in the excess-return regression.
- Assumptions: mean–variance investors, homogeneous expectations, borrow/lend at r_f, frictionless. Everyone holds the market (tangency) portfolio. Black zero-beta version replaces r_f with $E[R_z]$.
- APT: $R_i = a_i + \sum_j \beta_{ij} F_j + \epsilon_i \Rightarrow E[R_i] - r_f = \sum_j \beta_{ij} \lambda_j$ (no-arbitrage, approximate, factors unnamed). Negative loading = hedge, lowers expected return.

4.2 Dividend discount model

- Gordon: $P = D_0(1+g)/(r-g) = D_1/(r-g)$, $r > g$; very sensitive to $r - g$. Don't confuse D_0 and D_1.
- Two-stage DDM (high growth for n_h years, then g_l); implied growth $g = (Pr - D_0)/(D_0 + P)$.

4.3 Linear regression

- $\hat{\beta} = (X^\top X)^{-1} X^\top y$; $\text{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}$; hat matrix symmetric, idempotent. Gauss–Markov ⇒ BLUE.
- R^2, adjusted R^2, F-test, $t_j = \hat{\beta}_j / \widehat{se}_j$.
- Diagnostics: Breusch–Pagan/White (heteroscedasticity → HC errors), Durbin–Watson $\approx 2(1-\hat{\rho})$ (autocorrelation → Newey–West), VIF $= 1/(1-R_j^2) > 10$ (multicollinearity), Jarque–Bera (normality).
- statsmodels: `sm.OLS(y,X)` needs `add_constant`; formula API `ols('y~x')` adds an intercept.

4.4 Measuring signals: IC, IR, MI

- IC = (rank) correlation of signal with forward return, per period then averaged; $t \approx \text{IC}\sqrt{N}$. $|\text{IC}| < 0.02$ noise, > 0.05 strong.
- IR = alpha / tracking error; Fundamental Law IR $\approx \text{IC}\sqrt{\text{BR}}$.
- Mutual information $I(X;Y) = D_{KL}(p_{XY} \| p_X p_Y) = H(X) + H(Y) - H(X,Y) \geq 0$; $= 0$ iff independent; catches nonlinear dependence, no sign.

4.5 Storage: HDF5 vs. Parquet

- HDF5: groups, datasets, attributes, links; chunking needed for compression/resizing. HDFStore *fixed* (fast, read-all) vs. *table* (append, `where`). `where` only on `data_columns`/index; `min_itemsize` for strings.
- Parquet: columnar; row groups → column chunks → pages; footer min/max stats enable predicate pushdown; dictionary/RLE encodings; immutable files; Hive partitioning prunes directories.
- Column subsets and scans favor Parquet; indexed lookups of thin names favor HDF5 tables (unless Parquet sorted by key).
- ITCH: prices integer $\times 10^4$, ns timestamps, `stock_locate` from R messages. Repo hygiene: raw data never edited, logic in `src/`.

4.6 Minute bars (continues in Week 5)

- OHLCV, VWAP $= \sum p_i s_i / \sum s_i$, dollar volume, trade count; OFI $= n^{ask} - n^{bid}$.

5 Week 5: Strength of Signal; Smoothing Techniques

5.1 Minute bars and trade direction (cont'd from Week 4)

- VWAP is volume-weighted (300@50.00 + 100@50.20 ⇒ 50.05). Normalized OFI $\in [-1, 1]$.
- Trade sign from ITCH: join executions (E/C) to adds (A/F) by order reference; aggressor is contra to resting side. Tick rule otherwise.

5.2 Signal strength: IC, ICIR, MI (cont'd from Week 4)

- IC t-stat: $0.04\sqrt{2500} = 2$. ICIR = mean/std of IC_t; overlapping horizons inflate t.
- MI properties: symmetric, invariant to bijections, data processing inequality. $Y = X^2$: zero correlation, positive MI.
- NMI $= I/\sqrt{H(X)H(Y)} \in [0,1]$; binned MI has a positive noise floor $\approx (B-1)^2/(2n \ln B)$.
- Intraday signals: own order flow, SPY gap/residual, VWAP reversion $s_t = (C_t - \text{VWAP}_t)/\text{VWAP}_t$ (charge costs; don't trade at the signal's own close).

5.3 Kalman filter

- State space: $x_{t+1} = Ax_t + b + \epsilon^{(1)}$, $\epsilon^{(1)} \sim N(0, Q)$; $z_t = Cx_t + d + \epsilon^{(2)}$, $\epsilon^{(2)} \sim N(0, R)$.

- Predict: $\hat{x}_{t|t-1} = A\hat{x}_{t-1}$, $P_{t|t-1} = AP_{t-1}A^\top + Q$.
- Update: $K_t = P_{t|t-1}C^\top(CP_{t|t-1}C^\top + R)^{-1}$, $\hat{x}_t = \hat{x}_{t|t-1} + K_t(z_t - C\hat{x}_{t|t-1})$, $P_t = (I - K_tC)P_{t|t-1}$.
- Q/R sets the gain: large Q or small R tracks data; small Q or large R smooths. Adaptive, no window, $O(1)$ per step.
- Filter is causal (tradable); smoother uses future data (not tradable). EM estimates Q, R.

5.4 Roll model of the bid–ask spread

- $m_t = m_{t-1} + u_t$, $p_t = m_t + cq_t$, $q_t = \pm 1$: $\gamma_0 = \sigma_u^2 + 2c^2$, $\gamma_1 = -c^2$, $\gamma_{k\geq 2} = 0$.
- $\hat{c} = \sqrt{-\gamma_1}$, $\hat{\sigma}_u^2 = \gamma_0 + 2\gamma_1$; undefined if $\gamma_1 > 0$. Kalman form: $A = C = 1$, $Q = \sigma_u^2$, $R = c^2$.
- Correlated trade signs (one order → many fills) understate the spread: $\gamma_1 = -c^2(1-\rho)^2$.

5.5 Wavelet denoising

- Wavelets isolate time and frequency characteristics.
- Orthogonal transform preserves energy; i.i.d. noise stays σ^2 per coefficient; signals are sparse.
- Hard threshold $d\,\mathbf{1}\{|d| > \lambda\}$ (keeps jumps, rougher) vs. soft $\text{sign}(d)(|d| - \lambda)_+$ (smoother, biased).
- VisuShrink: $\hat{\sigma} = \text{median}(|d_1|)/0.6745$, $\lambda = \hat{\sigma}\sqrt{2\ln n}$; approximation coefficients untouched.
- Whole-day denoising is look-ahead; the causal version uses a rolling window and is weakest at the edge.

6 Week 6: The Alpha Hunt

6.1 Kalman applications (cont'd from Week 5)

- Time-varying beta: $R_{i,t} = \alpha_t + \beta_t R_{m,t} + e_t$, $\beta_{t+1} = \bar{\mu} + \phi(\beta_t - \bar{\mu}) + u_t$; state (β_t, α_t), $C_t = [R_{m,t}\ 1]$ (only C_t varies).
- Half-life $\ln(1/2)/\ln\phi$; stationary variance $\sigma_\beta^2/(1-\phi^2)$; $\phi \to 1$ is a random walk. Kalman beta is smoother than rolling OLS and shrinks toward $\bar{\mu}$.
- Futures curve (Schwartz–Smith): $\ln S_t = \chi_t + \xi_t$, short factor mean-reverting (half-life $\ln 2/\kappa$), long factor random walk with drift; $\ln F_t(T) = e^{-\kappa T}\chi_t + \xi_t + A(T)$.
- Missing quotes: drop rows of the observation matrix (never use stale prices); the filter infers the back contract from the front. Estimate by MLE on innovations.

6.2 Post-earnings-announcement drift (PEAD)

- SUE = (actual EPS − median forecast)/price (IBES); seasonal random-walk SUE from Compustat.
- After large positive (negative) surprises, prices drift up (down) for weeks; sort into SUE quintiles, measure CAR vs. market.
- Look-ahead fixes: correct information day (after-close → next day), only pre-announcement forecasts, PIT index membership, breakpoints from past data only. Restatements remain an issue.

6.3 Timing and backtest hygiene

- Signal formed at close of day d earns from $d+1$; never pair the day-d signal with the day-d return. After-close news is tradable next close.
- Include delisting returns; use a PIT universe; drop stories that merely recap prior returns.
- Report IC by horizon and IC decay, Newey–West t (lags $h-1$), overlapping portfolios (Jegadeesh–Titman), turnover, costs, Sharpe ($\times\sqrt{252}$), drawdown.

6.4 News sentiment: RavenPack and FinBERT

- RavenPack: filter on relevance, novelty, event detection; score ESS $\in [-1, 1]$; aggregate daily mean score and story count.
- FinBERT: 3 classes (pos/neg/neutral), score = $P(\text{pos}) - P(\text{neg}) \in [-1, 1]$.
- Fine-tuning on abnormal-return terciles: label windows must end before the validation/test period; select epoch by validation IC (overfitting after a few epochs).

Recurring themes. Look-ahead/survivorship bias (rolls, corporate actions, PIT universes, COT release dates, news timing, smoothers); returns are data; honest signal statistics (overlapping data, MI noise floors). **Look-ahead data can creep into your backtests in unexpected ways!**